THE QUOTIENT STACK OF A SMOOTH GROUPOID IN ALGEBRAIC SPACES IS AN ALGEBRAIC STACK

THE STACKS PROJECT AUTHORS

Editorial note. Condition (1) is 'by construction', anchored on the quotient stack construction; carried as an explicit obligation. The inputs not carded are declared external.

Allowed standard prerequisites (editorial).

- Diagonal of quotient stack. lemma-diagonal-quotient-stack (same chapter, NOT carded, declared external). Condition (2), in the stronger form.
- Presheaf transfer of etale. lemma-map-presheaves-representable-by-spaces-transformation-property (same chapter, NOT carded, declared external). S_W -> S_U is surjective etale.
- Property implication. lemma-representable-transformations-property-implication (same chapter, NOT carded, declared external). Etale implies smooth here.
- Smooth presentation of the quotient. lemma-smooth-quotient-smooth-presentation (carded L26692). S_U -> [U/R] is surjective smooth.
- Composition property transfer. lemma-composition-representable-transformations-property (same chapter, NOT carded, declared external). Composes S_W -> S_U with S_U -> [U/R].

Source reference. Original definition 026O: sources/stacks-algebraic/pinned-original-source.tex, source lines 1506–1506. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 045B: sources/stacks-algebraic/pinned-original-source.tex, source lines 1159–1159. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04WZ: sources/stacks-algebraic/pinned-original-source.tex, source lines 2571–2571. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 045A: sources/stacks-algebraic/pinned-original-source.tex, source lines 1118–1118. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 045F: sources/stacks-algebraic/pinned-original-source.tex, source lines 1264–1264. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04X0: sources/stacks-algebraic/pinned-original-source.tex, source lines 2586–2586. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Theorem 0.1. *Let S be a scheme contained in Sch_{fppf}. Let (U, R, s, t, c) be a smooth groupoid in algebraic spaces over S. Then the quotient stack $[U/R]$ is an algebraic stack over S.*

Proof. We check the three conditions of Definition 026O. By construction we have that $[U/R]$ is a stack in groupoids which is the first condition.

The second condition follows from the stronger Lemma 04WZ.

Finally, we have to show there exists a scheme W over S and a surjective smooth 1-morphism $(Sch/W)_{fppf} \longrightarrow \mathcal{X}$. First choose $W \in \mathrm{Ob}((Sch/S)_{fppf})$ and a surjective étale morphism $W \to U$. Note that this gives a surjective étale morphism $\mathcal{S}_W \to \mathcal{S}_U$ of categories fibred in sets, see Lemma 045A. Of course then $\mathcal{S}_W \to \mathcal{S}_U$ is also surjective and smooth, see Lemma 045F. Hence $\mathcal{S}_W \to \mathcal{S}_U \to [U/R]$ is surjective and smooth by a combination of Lemmas 04X0 and 045B. □